

Mobile Customers Data Anonymisation Report

Name: [Harshit Shukla]

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Introduction

In this project, I worked on anonymising customer data provided in the file `mobile_customers.xlsx`. The dataset contained sensitive customer information collected from users who signed up for the mobile application over the last three years.

The main objective of this task was to protect customer privacy by anonymising personal and sensitive information while still preserving useful data for analysis by the data science team at InsightSpark.

I completed this task using Python in Google Colab with the help of the Pandas library.

Tools and Technologies Used

For this project, I used the following tools and technologies:

- Python
- Pandas Library
- Google Colab
- Microsoft Excel

These tools helped me efficiently load, clean, transform, anonymise, and export the dataset.

Understanding the Dataset

Before starting the anonymisation process, I explored the dataset to understand its structure and identify sensitive columns.

The dataset included information such as:

- Customer IDs
- Usernames
- Names
- Addresses
- Email addresses
- Birthdates
- Gender
- Current locations
- Job information
- Salary information
- Credit card details

I used Pandas functions such as `head()` and `columns` to inspect the dataset and identify which fields contained personally identifiable or sensitive information.

Data Anonymisation Techniques Used

To anonymise the dataset while preserving its analytical value, I applied multiple anonymisation techniques.

1. Removing Sensitive Columns

I removed highly sensitive columns that were not necessary for analysis, including:

- Name
- Address
- Birthdate
- Credit card number
- Credit card security code
- Credit card expiry information

These fields could directly identify customers or expose sensitive financial information.

2. Masking Personal Information

Some columns were partially masked instead of being removed completely.

Customer ID Masking

Customer IDs were partially masked to hide the original identifiers while still maintaining uniqueness.

Username Masking

Username were masked by keeping only the first few characters and hiding the remaining portion.

Email Masking

Email addresses were anonymised by partially masking the username and domain information.

This approach helped preserve useful patterns in the data without exposing personal identities.

3. Categorising Numerical Data

Instead of keeping exact personal values, I converted them into broader categories.

Age Grouping

Exact ages were replaced with age groups such as:

- 19–30
- 31–45
- 46–60

This helped reduce the risk of identifying individuals while preserving demographic information.

Salary Bracketing

Exact salary values were converted into salary brackets such as:

- Low
- Medium
- High
- Very High

This maintained useful income trends for analysis while protecting financial privacy.

Final Dataset

After anonymisation, the final dataset retained useful analytical information such as:

- Gender
- Location
- Employer
- Job role
- Age groups
- Salary brackets
- Credit card provider

At the same time, all direct personal identifiers and sensitive financial details were removed or masked.

The anonymised dataset was then exported as a CSV file for submission.

Conclusion

I successfully anonymised the mobile customer dataset using Python and Pandas while preserving the usefulness of the data for future analysis.

Through this project, I learned how to apply practical data privacy techniques such as:

- Data masking
- Data removal
- Data categorisation
- Data transformation
- Privacy-focused data handling

This project gave me practical experience in handling sensitive customer information and preparing anonymised datasets for secure business and analytical use.